

1. What is Data Science? Explain its primary objective.

Data Science is the process of collecting, analyzing, and interpreting data using scientific methods, algorithms, and machine learning.

Primary Objective: To extract meaningful insights from data for better decision-making.

2. Why is Data Science important in today's digital world? Mention any four reasons.

- Helps in data-driven decision making.
- Predicts future trends.
- Improves business efficiency.
- Supports automation and innovation.

3. List any four benefits of Data Science in business and society.

- Better decision-making.
- Improved customer satisfaction.
- Fraud detection.
- Healthcare and disease prediction.

4. Mention four real-world applications of Data Science.

- Healthcare
- Banking and Finance
- E-commerce
- Transportation

5. What are the different facets of data? Briefly explain each.

- **Structured Data:** Organized in tables (e.g., databases).
- **Semi-Structured Data:** Has partial organization (e.g., XML, JSON).
- **Unstructured Data:** No fixed format (e.g., images, videos, text).

6. Differentiate between structured, semi-structured, and unstructured data with suitable examples.

Type	Description	Example
Structured	Fixed format	Excel, SQL tables
Semi-Structured	Partial structure	XML, JSON
Unstructured	No predefined format	Images, Videos, Emails

7. What is Big Data? State its key characteristics (5 Vs).

Big Data refers to extremely large datasets that cannot be processed using traditional methods.

5 Vs:

- Volume
- Velocity
- Variety
- Veracity
- Value

8. Explain the Big Data ecosystem. Name any four technologies used in it.

The Big Data ecosystem consists of tools used for storing, processing, and analyzing large datasets.

Technologies:

- Hadoop
- Spark
- Hive
- HBase

9. What are the major phases of the Data Science process?

Major Phases of Data Science process are

- Data Collection
- Data Cleaning
- Data Analysis
- Model Building

- Model Evaluation
- Deployment
- Visualization

10. What is data retrieval? Mention two common sources from which data can be retrieved.

Data retrieval is the process of obtaining data from various sources.

Sources:

- Databases
- APIs

11. Why is data collection considered an important step in the Data Science process?

Because accurate and relevant data leads to reliable analysis and better model performance.

12. What is data cleansing? List any four common data-cleaning techniques.

Data cleansing is the process of correcting or removing inaccurate data.

Techniques:

- Remove duplicate records.
- Handle missing values.
- Correct inconsistent data.
- Remove outliers.

13. What is data integration? Why is it necessary in Data Science?

Data integration combines data from multiple sources into one dataset.

Need:

- Improves data quality.
- Provides a complete view for analysis.

14. Explain data transformation with suitable examples.

Data transformation converts data into a suitable format for analysis.

Examples:

- Converting text to numbers.
- Normalization of data.
- Date format conversion.

15. What is data analysis? Mention any four techniques used for data analysis.

Data analysis is the process of examining data to extract useful information.

Techniques:

- Statistical Analysis
- Regression Analysis
- Classification
- Clustering

16. Differentiate between descriptive, predictive, and prescriptive analytics.

Analytics	Purpose
Descriptive	What happened?
Predictive	What will happen?
Prescriptive	What should be done?

17. What is meant by building a model in Data Science? Mention the basic steps involved.

Model building is the process of training an algorithm to make predictions.

Steps:

- Select algorithm.
- Split data.
- Train model.
- Test model.

- Tune parameters.

18. Why is model evaluation important? Name any four evaluation metrics used for classification models.

Model evaluation checks the performance and accuracy of the model.

Metrics:

- Accuracy
- Precision
- Recall
- F1-Score

19. Why is presenting findings an essential part of the Data Science process? Mention any four visualization tools or techniques.

It helps stakeholders understand insights and make informed decisions.

Tools/Techniques:

- Bar Chart
- Line Chart
- Pie Chart
- Tableau

20. How are Data Science models deployed into real-world applications? Give any four examples of Data Science applications.

Models are deployed using web applications, cloud platforms, mobile apps, or APIs so users can make real-time predictions.

Applications:

- Recommendation Systems (Netflix, Amazon)
- Fraud Detection
- Disease Prediction
- Weather Forecasting